

Yavor Litchev

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COMPUTER LANGUAGES

Java, Python, C, C++, Rust, Arduino, OpenSCAD, HTML, CSS, and JavaScript.

OTHER COMPUTER SCIENCE COURSES & SKILLS

- **Practicum in Artificial Intelligence**, KByte Computer Science Academy
- **Deep Learning**, KByte Linux tools, CUDA + GPU accelerated Python research environment, Tensorflow-GPU 1.4, Keras 2.1.4 - tools compiled with lubcudnn 6 and 7. Topics: deep networks, single shot detection, convolutional or vectorized models for language, demo projects featuring AlphaZero Go and GAN inspired sequence to sequence learning. Application of neural networks to text, images, and other data, practice parsing data and preparing it for use on research servers.
- **Statistical Learning**, KByte This course is taught in Python using the numpy and sk-Learn libraries. (Vectors and Dot Product; Intro to Stats, Conditional Probability, Bayes Theorem, Naive Bayes Classifier; Central Limit Theorem, Text Classification with Naive Bayes Classifier; Linear Regression; Gradient Descent; Logistic Regression; Neural Networks; Q Learning, Principal Component Analysis; Clustering)
- **Intermediate Algorithms**, KByte, AP level course (Recursion, Stacks and Queues, Floodfill, Selection Sort, Binary Search, Merge Sort, Quick Sort, Objects and References, Linked List, Linked Lists Algorithms, Games with Linked Lists)
- **Problem Solving with Java**, KByte (N-Dimensional Arrays, Recursion, Hanoi, Flood Fill, Classes, ArrayList & References)
- **USACO Gold prep course**, KByte, (Graph Search; HashSet and TreeSet; Sliding Window, Combinatorics; Dynamic Programming; Priority Queues; Coordinate Compression; Union-Find Data Structure and Minimal Spanning Tree; Binary Search; $O(\log(n))$ Algorithms; Binary Index Tree, etc.)

EDUCATION

Stanford University

- **MS in Computer & Network Security**, GPA 4.05/4, (09/2023-2027)
- **BS in Computer Science (Individually designed track - Cybersecurity & Cryptography)** GPA 3.94/4 (09/2022-04/2026)
- **Coursework:**
 - Advanced Topics in Cryptography (CS 355)
 - Introduction to Cryptography (CS 255)
 - Cryptocurrencies & Blockchain Technologies (CS 251)
 - Design & Analysis of Algorithms (CS161)
 - Number Theory for Cryptography (MATH 110)
 - Quantum Computing (CS 259Q)
 - Convex Optimization II (EE 364B)
 - Performance Engineering of Computer Systems & Networks (EE 384S)
 - Program Analysis & Optimizations (CS 243)
 - Programming Languages (CS 242)
 - Introduction to Probability Theory (MATH 151)
 - Parallel Computing (CS 149)
 - Infrastructure at Scale (SC 153)
 - Data Management & Data Systems (CS 145)
 - Linear Algebra & Multivar. Calculus (MATH 51)
 - Topics in Computer & Network Security (CS 356)
 - Computer & Network Security (CS 155)
 - Formal Methods for Computer Systems (CS 357S)
 - Advanced Operating Systems Lab (CS 340LX)
 - Advanced Systems Laboratory (CS 240 LX)
 - Operating Syst. Design & Implementation (CS 140E)
 - Operating Systems Principles (CS 111)
 - Computer Organization & Systems (CS 107)
 - Internet Switching & Routing Protocols (EE 384A)
 - Introduction to Internet of Things (EE 284A)
 - Internet Principles & Protocols Data (EE 184)
 - Introduction to Computer Networking (CS 144)
 - Machine Learning (CS 229)
 - Deep Learning for Computer Vision (CS 231N)
 - Circuits I (EE 101A)
 - Mathematical Foundations of Computing (CS 103)
 - Differential Equations with Linear Algebra (MATH 53)

RESEARCH EXPERIENCE

- **Intern, Cryptography Research Team, Ethereum Foundation** 06/2026-09/2026
Developed an AI-assisted zkVM review pipeline utilizing a generalized execution relation (R_{exec}) to generate and manually audit security-proof roadmaps for six major submissions (e.g. SP1, ZisK). Formalized VanillaVM's Correct-Trace Extractability (CTE) theorem in Lean 4, implementing a memory-reconstruction layer, a 5-class representative ISA, and an injective segment-bus resolution to mathematically guarantee full-memory trace validity from any accepted proof.
- **Research Assistant to Prof. Dan Boneh** 03/2026-06/2026
We sought to develop a threshold digital signature scheme that is plausibly post quantum secure. We wished for the construction to be able to use hash functions, as the hardness assumptions for hash functions are more widely accepted in the Bitcoin community as opposed to lattice assumptions. We eventually created a scheme with a keyed PRF function that may use either hashes or lattices in its implementation, and the final signature is identical in size to that of a single person's signature and can be verified with existing Bitcoin opcodes/verification logic.
- **Senior research project with advisor Prof. Dan Boneh** 09/2025-12/2025
Developed YavOracle, a Rust framework that enforces coverage-safe Fiat-Shamir transcripts at compile time. <https://rovayl.github.io/YavOracle/>
- **Machine Learning course individual project / research tooling** 10/2025-12/2025
Developed a zk-underconstraint detector that analyzes R1CS circuits for structural bugs. <https://github.com/RovayL/zk-underconstraint-detector>
- **CURIS Research Intern, Stanford University Computer Science Department** 06/2024 - 08/2024
Process isolation is critical for the security of computer systems. Hardware protections are typically used, but can be slow when changing protection domains. Software-based Fault Isolation (SFI) solves this issue, and Lightweight Fault Isolation (LFI), the basis of this research project, is a versatile and performance orientated example of this. This project sought to extend the already highly versatile Lightweight Fault Isolation (LFI) system, and we were able to make the LFI system more efficient in verification, more flexible in which Instruction Set Architecture (ISA) can be used, and more able to accommodate changes in ISA architectures.
- **Student Researcher, MIT PRIMES, Massachusetts Institute of Technology** 01/2020 - 01/2022
Research in cryptography, machine learning and artificial intelligence. Developed a hybrid privacy scheme, which combined local differential privacy (LDP) and multiparty computation (MPC) to optimize secure learning in vector machine training. My partner

worked on the LDP portion of the project and I focused on the MCP part and later on combining the two. The created hybrid model is more accurate than a pure LDP and much faster than a pure MPC model. (2020)

Research in cryptographic digital signatures. Created a multi-signature scheme with a monotonic access pattern, which allows for greater degree of access control, while preserving security and correctness. (2021)

TEACHING EXPERIENCE

- **Teaching Assistant for CS 251 Cryptocurrencies & Blockchain Technologies** (10/2026-12/2026)
- **Course Assistant for CS 255 Introduction to Cryptography** (01/2026-03/2026)
Helped students during office hours, created autograders for various homework assignments, graded students' homework, and thoroughly responded to students' questions on EdStem in a timely and professional manner.
- **Tutor for Math 51 Linear Algebra, Multivariable Calculus, and Modern Applications and Math 53 Differential Equations with Linear Algebra, Fourier Methods, and Modern Applications at the Stanford Center for Teaching and Learning** (09/2023- 05/2025)
Helped students clarify class concepts, mathematical questions, and also generally how students should best prepare in order to succeed in the class, whether or not it be explaining how to approach and understand homework question to providing studying strategies for exams.
- **Computer Science Teaching Assistant, KTBYTE Computer Science Academy** 2/2018 - 6/2022
In person, online and office hours teaching assistant helping students to understand and apply computer science concepts to solve class and homework problems.
- **Robotics Club Leader / Instructor, KTBYTE Computer Science Academy** 6/2019 - 8/2022
Taught the Arduino Microcontroller and its applications. Curriculum development, instruction, evaluation, communications with students and parents.

PROFESSIONAL EXPERIENCE

- **Cryptographic/Security Engineer, XTrace Inc. (part-time)** 1/ 2025 - Current
The company performs vector search on encrypted ciphertext in order to give private AI inference, and I worked on acceleration on said vector embedding search over encrypted ciphertext via partially homomorphic schemes. The primary scheme used is the Paillier encryption scheme, from which alternative partial homomorphic encryption schemes are implemented and benchmarked, and code designed for various hardware accelerators and optimized encryption techniques (AVX512, CUDA for GPUs, lookup tables, reduced ring sizes, etc.) was used to further accelerate the Paillier Encryption Scheme.
- **Software Developer Intern, Innoiso, San Francisco, CA** 6/2023 - 8/2023
Developed a program, which automates website creation given specific database structure. The application allows small businesses to create quickly and cost effectively a custom website with front-end/back-end integration. A GUI application, which utilizes HTML, CSS, Java Script and Vue.js enables a user to select what DB structure to create, after which the GUI stores the template information in a separate JSON file. A second Python program where the frontend, backend, SQL, and PHP portions of the web application are all constructed, can then be executed to launch the completed website.

PUBLICATIONS & PRESENTATIONS

- **"Batched Paillier-Based Hamming-Distance Computation over Binary Embeddings"**, Litchev, Y., Ouyang, L., <https://arxiv.org/abs/2609.21364>, Sept 18, 2026
- **"Expanding LFI Machine Code Validation with Decision Diagrams"**
Litchev, Y., Mentors: Yedidia, Z., Mazières, D., CURIS 2024 Poster Presentation, Stanford University, Aug 29, 2024
- **"Hybrid Privacy Scheme"**
Litchev, Y., Thomas, A., 2nd AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI-21) Poster Session Presentation, Feb 8-9, 2021 (<https://ppai21.github.io/files/28-paper.pdf>)
- **"Signature Scheme with Access Control"**,
Litchev, Y., American Mathematical Society-Pi Mu Epsilon (AMS-PME) Undergraduate Student Poster Session, 2022 Joint Mathematics Meetings (<https://meetings.ams.org/math/jmm2022/meetingapp.cgi/Paper/11069>)

PROJECTS

- **Decentralized Cryptocurrency Exchange Implementation in Solidity** 10/2023-12/2023
Cryptocurrencies and Blockchain Technologies course individual project
Created an ERC20 token (traded over the exchange) while ensuring the code is resilient against adversarial attacks. Created and deployed a decentralized exchange (DEX), e.g., an automated market maker (AMM), modeled after Uniswap V1. Built JavaScript backend. OpenZeppelin ERC-20 contract implementation. Exercised the contracts on an Ethereum testnet (deploying, adding/removing liquidity, executing swaps, and inspecting state through the UI and a block explorer). Built a user interface that computes useful information for the user and allows non-programmers to use the Dapp

LEADERSHIP AND PROFESSIONAL INVOLVEMENT

- **Stanford Blockchain Club - Vice President - Finance** June 2025 - Current
- **Stanford Blockchain Club - Cryptography Research Lead, Co-host, Host, Contributor to Stanford Blockchain Review** Oct 2024 - Current
Interview professors and cryptographic researchers on their recent work. Blogposts include recorded video interviews accompanied by articles that explain the contributions and their effect on the Blockchain ecosystem and privacy preserving solutions in general.
 - [#76 - Cryptography Research Spotlight - An Overview of the LatticeFold Architectural Family](#)
A Conversation with Prof. Dan Boneh of Stanford University, *Stanford Blockchain Club*, Oct 14, 2025, *Stanford Blockchain Review*, Vol 8, Article No. 6, Litchev, Y.
 - [#74 - Cryptography Research Spotlight — zkTLS and the DECO Protocol](#)
A Conversation with Prof. Fan Zhang of Yale University, *Stanford Blockchain Club*, Aug 13, 2025, *Stanford Blockchain Review*, Vol 8, Article No. 4, Hosts: Yu, J., Pantera Capital, Litchev, Y., SBC
 - [#71 - Cryptography Research Spotlight - An Architectural Overview of RISC Zero](#)
A Conversation with Jeremy Bruestle, CEO and Cofounder of RISC Zero, *Stanford Blockchain Club*, May 16, 2025, *Stanford Blockchain Review*, Vol 8, Article No.1, Litchev, Y., Yu, J.
 - [#53 - Cryptography Research Spotlight: BPR Traitor Tracing for Threshold Decryption](#)
A Conversation with Prof. Dan Boneh of Stanford University, *Stanford Blockchain Club*, Oct 16, 2024, *Stanford Blockchain Review*, Volume 6, Article No. 3, Yu, J., Litchev, Y.
- **IC3 (The Initiative for Crypto Currencies and Contracts) Blockchain Camp 2025**, June 9-15, 2025 Cornell Tech (Roosevelt Island) New York, NY.
Member of a Team, lead by Prof. Fan Zhang (Yale University, IC3) which placed 3rd in the Hackathon. The team built a system for bug hunters to responsibly disclose smart contract bugs to deployers with confidentiality and spam prevention. DeadDrop uses Oblivious Message Retrieval (OMR) and TEEs to ensure private communication between bug hunters and smart contract developers.